CFD and AI-Based Hydrogen Leak Localization

in a 3D Indoor Environment

Towards Autonomous Robotic Inspection Systems

Independent Research Project

Mhamad Al Sheikh

Physics Researcher

Lebanon

May 2026

Abstract

Hydrogen is expected to play a major role in future clean-energy systems. However, due to its low density, high diffusivity, and wide flammability range, accidental hydrogen leaks may represent a significant safety risk in industrial facilities, storage infrastructures, and enclosed environments.

This project presents a computational framework for hydrogen leak localization in a three-dimensional indoor environment by combining Computational Fluid Dynamics (CFD), sensor networks, and Artificial Intelligence (AI). The study was conducted using OpenFOAM to simulate hydrogen dispersion under multiple leak scenarios inside a ventilated room. A network of fixed sensors was deployed to monitor hydrogen concentration evolution over time.

A total of 106 leak scenarios were generated, including different leak locations and release conditions. The simulation results were processed to create a dataset containing sensor responses associated with known leak coordinates. Machine learning techniques were then employed to estimate the leak position from sensor measurements. A Random Forest regression model was selected due to its robustness and ability to capture nonlinear relationships between sensor observations and leak locations.

The generated dataset contains approximately 1.59 million sensor measurements and 106 independent leak scenarios. The trained model achieved a localization accuracy characterized by a Root Mean Square Error (RMSE) of approximately 0.65 m along the x-axis and 0.15 m along the y-axis. Furthermore, a sensor-ranking analysis was performed to identify the most influential sensors for leak localization. The results indicate that ceiling-mounted sensors provide the highest information content due to the natural buoyancy of hydrogen.

The proposed framework demonstrates the feasibility of combining CFD simulations and machine learning for hydrogen leak localization and establishes a foundation for future developments involving autonomous robotic inspection systems and real-time industrial safety monitoring.

Contents

Acknowledgements

This work was carried out as an independent personal research project.

The author would like to acknowledge the developers of OpenFOAM, ParaView, Python, and Scikit-Learn for providing powerful open-source tools that made this study possible.

Special appreciation is also extended to the scientific open-source community for enabling advanced research through freely accessible computational resources.

Chapter 1

Introduction

1.1 Background

Hydrogen is considered one of the most promising energy carriers for future clean-energy systems. It can be used in transportation, power generation, industrial processes, and energy storage. However, hydrogen also introduces important safety challenges because it is colorless, odorless, highly diffusive, and flammable.

When hydrogen leaks inside an enclosed environment, it rises rapidly because of its very low density compared with air. This behavior may lead to accumulation near the ceiling, especially in poorly ventilated or confined regions. Therefore, hydrogen safety requires not only detection systems, but also intelligent methods capable of estimating the leak source location.

1.2 Problem Statement

Traditional hydrogen leak detection systems generally rely on fixed gas sensors distributed throughout the monitored environment. These sensors can detect the presence of hydrogen when the local concentration exceeds a certain threshold. However, they do not always provide enough information to identify the exact position of the leak source.

The main challenge addressed in this project is therefore the localization of hydrogen leaks in a three-dimensional indoor environment using sensor measurements generated from CFD simulations.

1.3 Motivation

Computational Fluid Dynamics provides a powerful tool to simulate hydrogen dispersion under controlled conditions. It allows the visualization of plume evolution, ceiling accumulation, ventilation effects, and sensor response.

Artificial Intelligence can then be used to learn the relationship between sensor measurements and leak positions. By combining CFD-generated datasets with machine-learning models, it becomes possible to estimate leak coordinates from sensor readings.

This combination of CFD and AI represents a promising step toward intelligent hydrogen safety systems and future autonomous robotic leak detection platforms.

1.4 Project Objectives

The objectives of this work are:

- Simulate hydrogen dispersion in a 3D indoor environment.
- Study buoyancy-driven hydrogen transport.
- Design and evaluate a fixed sensor network.
- Generate a CFD-based dataset for machine learning.
- Train an AI model for hydrogen leak localization.
- Identify the most informative sensor positions.
- Establish a foundation for future autonomous robotic hydrogen leak detection.

1.5 Report Organization

This report is organized as follows.

Chapter 2 presents the physical background of hydrogen dispersion.

Chapter 3 describes the CFD methodology, computational domain, and simulation setup.

Chapter 4 introduces the sensor network architecture and data acquisition process.

Chapter 5 presents the CFD simulation results and discusses hydrogen plume evolution.

Chapter 6 describes the dataset generation process and the Artificial Intelligence framework used for leak localization.

Chapter 7 presents the sensor ranking analysis and feature importance evaluation.

Chapter 8 introduces a future autonomous robotic inspection system based on the developed framework.

Finally, Chapter 9 summarizes the main findings and discusses future research directions.

Chapter 2

Physical Background

2.1 Hydrogen Properties

Hydrogen is the lightest element in the periodic table and possesses several unique physical properties that strongly influence its behavior after a leak occurs. Compared with air, hydrogen has a significantly lower density, which causes it to rise rapidly under the influence of buoyancy.

In addition to its low density, hydrogen exhibits a high diffusion coefficient, allowing it to spread quickly within enclosed environments. Hydrogen is also colorless and odorless, making accidental releases difficult to detect without dedicated monitoring systems.

These characteristics make hydrogen an attractive energy carrier but also introduce important safety challenges that require efficient detection and localization methods.

2.2 Hydrogen Dispersion Mechanisms

After a leak occurs, hydrogen transport is governed by several physical mechanisms acting simultaneously.

The first mechanism is advection, which corresponds to the transport of hydrogen by the surrounding airflow. Ventilation systems and natural air circulation can significantly influence the direction and speed of plume propagation.

The second mechanism is molecular diffusion. Diffusion tends to smooth concentration gradients and promotes the mixing of hydrogen with the surrounding air.

The third mechanism is buoyancy, which plays a dominant role in hydrogen dispersion because hydrogen is much lighter than air.

The combined effect of these mechanisms determines the shape, evolution, and final distribution of the hydrogen plume.

2.3 Buoyancy Effects

Buoyancy is one of the most important phenomena governing hydrogen transport in enclosed environments.

Because hydrogen has a lower density than air, the released gas experiences an upward force that causes it to rise rapidly. As the plume reaches the ceiling, it begins to spread horizontally and forms an accumulation layer in the upper region of the room.

This physical behavior has important implications for sensor placement strategies. Sensors installed near the ceiling are generally exposed to hydrogen earlier than sensors located at lower elevations.

The CFD simulations performed in this project clearly demonstrate this phenomenon and confirm the importance of ceiling-mounted monitoring systems.

2.4 Ventilation Effects

Ventilation can significantly modify hydrogen dispersion patterns.

Air inlets introduce fresh air into the room, while outlets remove contaminated air. Depending on their positions and operating conditions, ventilation systems may either enhance hydrogen removal or redistribute the plume toward different regions of the enclosure.

Understanding the interaction between buoyancy and ventilation is essential for designing effective hydrogen monitoring systems.

2.5 Hydrogen Accumulation Risks

Hydrogen accumulation inside enclosed environments represents a potential safety hazard. If hydrogen concentration reaches flammable levels, accidental ignition may lead to fire or explosion events.

For this reason, early detection and localization of hydrogen leaks are critical for ensuring operational safety.

The development of intelligent monitoring systems capable of identifying leak locations rapidly can significantly reduce response time and improve risk mitigation strategies.

2.6 Sensor-Based Monitoring Systems

Modern hydrogen safety systems commonly rely on distributed sensor networks.

A sensor network provides spatial information about hydrogen concentration and allows continuous monitoring of the environment. When combined with advanced data analysis techniques, sensor measurements can be used not only to detect hydrogen but also to estimate the position of the leak source.

The integration of CFD simulations, sensor networks, and Artificial Intelligence forms the central concept investigated in this project.

Chapter 3

CFD Methodology

3.1 Simulation Software

The numerical simulations performed in this study were carried out using OpenFOAM v2412, an open-source Computational Fluid Dynamics platform widely used for research and industrial applications.

ParaView was employed for post-processing and visualization of hydrogen dispersion patterns, while Python was used for data extraction, dataset generation, statistical analysis, and machine-learning applications.

The complete workflow combines CFD simulations, sensor data extraction, and Artificial Intelligence analysis within a unified computational framework.

3.2 Computational Domain

The computational domain represents a three-dimensional indoor environment designed to investigate hydrogen dispersion following accidental leaks.

The simulated enclosure contains:

- A hydrogen leak source.
- An air inlet.
- An air outlet.
- A ceiling-mounted sensor network.
- Additional monitoring sensors near the outlet region.
- A reference sensor near the floor.

The geometry was selected to reproduce a realistic indoor environment where buoyancy and ventilation simultaneously influence hydrogen transport.

3.3 Hydrogen Leak Scenarios

To ensure sufficient variability for machine-learning applications, multiple leak configurations were simulated.

The study includes:

- Small leak scenarios.
- Medium leak scenarios.
- Large leak scenarios.
- Central leak positions.
- Corner leak positions.
- Random leak locations.

A total of 106 independent leak scenarios were generated and analyzed.

This large collection of simulations forms the basis of the AI dataset used later in the project.

3.4 Hydrogen Transport Equation

Hydrogen concentration was modeled using a scalar transport formulation.

The governing equation is:

$$\frac{\partial C}{\partial t} + \nabla \cdot (\mathbf{U}C) = \nabla \cdot (D \nabla C) + S \tag{3.1}$$

where:

- C is the hydrogen concentration.
- $\mathbf{U}$ is the airflow velocity field.
- D is the diffusion coefficient.
- S is the source term representing the leak.

This equation describes the combined effects of advection, diffusion, and hydrogen injection.

3.5 Physical Mechanisms Considered

Several important physical phenomena influence hydrogen transport inside the enclosure.

3.5.1 Advection

Hydrogen is transported by airflow generated by the ventilation system. The resulting velocity field determines the overall direction of plume motion.

3.5.2 Diffusion

Molecular diffusion promotes mixing between hydrogen and air and reduces concentration gradients over time.

3.5.3 Buoyancy

Because hydrogen is lighter than air, buoyancy causes the plume to rise rapidly toward the ceiling. This effect dominates the dispersion process in many scenarios.

3.5.4 Ventilation

Air inlet and outlet conditions modify plume evolution and contribute to hydrogen removal from the room.

3.6 Simulation Workflow

The simulation procedure consists of the following steps:

1. Geometry preparation.
2. Mesh generation.
3. Definition of boundary conditions.
4. Initialization of hydrogen source terms.
5. OpenFOAM simulation.
6. Sensor data extraction.
7. Dataset generation.
8. Machine-learning analysis.

The workflow allows the automatic generation of large quantities of simulation data suitable for Artificial Intelligence applications.

3.7 Expected Hydrogen Behavior

Based on the physical properties of hydrogen, the expected plume evolution follows several stages:

1. Initial release near the source.
2. Rapid upward motion due to buoyancy.
3. Ceiling accumulation.
4. Horizontal spreading near the ceiling.

5. Interaction with ventilation.

6. Quasi-steady distribution.

The CFD results presented later in this report confirm this expected behavior.

Chapter 4

Sensor Network Design

4.1 Introduction

A sensor network was designed to monitor hydrogen concentration throughout the simulated environment. The primary objective of the network is to provide sufficient spatial information for hydrogen leak detection and localization while maintaining a realistic monitoring strategy suitable for future industrial deployment.

The sensor measurements generated during the CFD simulations constitute the foundation of the machine-learning dataset developed in this work.

4.2 Sensor Network Architecture

The monitoring system consists of twenty-five fixed sensors distributed throughout the room.

The network includes:

- Twenty ceiling-mounted sensors.
- Four outlet monitoring sensors.
- One reference sensor near the floor.

The total number of sensors is therefore equal to twenty-five.

Sensor Type	Number
Ceiling Sensors	20
Outlet Sensors	4
Reference Floor Sensor	1
Total	25

Table 4.1: Sensor network configuration.

4.3 Ceiling Sensor Grid

The ceiling sensor grid represents the core of the monitoring system.

Hydrogen is significantly lighter than air and therefore tends to rise rapidly after release. As a result, elevated hydrogen concentrations are expected near the ceiling.

To exploit this behavior, twenty sensors were arranged in a regular ceiling-mounted grid covering the upper region of the room.

The grid configuration allows continuous monitoring of hydrogen accumulation and provides detailed spatial information regarding plume evolution.

4.4 Outlet Monitoring Sensors

Additional sensors were installed near the outlet region.

These sensors provide information regarding hydrogen removal by the ventilation system and help characterize plume transport toward the exhaust area.

Although outlet sensors generally detect hydrogen later than ceiling sensors, they remain useful for understanding global dispersion patterns.

4.5 Reference Floor Sensor

A reference sensor was positioned near the floor.

Because hydrogen rises due to buoyancy, the floor sensor is expected to experience lower concentrations than the ceiling sensors.

The reference sensor provides an additional indicator of vertical concentration gradients inside the enclosure.

4.6 Sensor Data Acquisition

During each CFD simulation, hydrogen concentration values were sampled at every sensor location.

Measurements were recorded throughout the simulation time in order to capture:

- Initial detection events.
- Concentration evolution.
- Peak concentration values.
- Detection times.
- Temporal dispersion dynamics.

The resulting time-series data were stored automatically for later processing.

4.7 Generated Database

The sensor network produced a large dataset suitable for machine-learning applications.

The final database contains:

- 106 independent leak scenarios.
- Approximately 1.59 million sensor measurements.
- Sensor response histories for all monitoring locations.
- Summary statistics for each leak scenario.

This database forms the basis of the Artificial Intelligence framework developed later in this report.

4.8 Sensor Performance Indicators

Several performance indicators were used to evaluate sensor effectiveness.

These indicators include:

- Detection coverage.
- Detection time.
- Maximum recorded concentration.
- Number of first detections.
- Distance between sensor and leak source.

These metrics allow the identification of the most informative sensors for hydrogen leak localization.

4.9 Importance of Sensor Placement

Sensor placement plays a critical role in leak detection performance.

The CFD simulations indicate that hydrogen accumulates predominantly near the ceiling due to buoyancy effects. Consequently, sensors installed in the upper region of the room generally outperform sensors located at lower elevations.

This observation is confirmed later by both the sensor ranking analysis and the machine-learning feature importance results.

4.10 Summary

The sensor network developed in this work provides comprehensive spatial coverage of the indoor environment and generates a large quantity of data suitable for Artificial Intelligence applications.

The combination of ceiling sensors, outlet sensors, and a reference floor sensor enables detailed monitoring of hydrogen dispersion and forms the foundation of the leak localization framework presented in the following chapters.

Chapter 5

CFD Results

5.1 Introduction

This chapter presents the Computational Fluid Dynamics results obtained from the hydrogen leak simulations. The objective is to investigate the evolution of hydrogen dispersion inside the room and identify the dominant physical mechanisms governing plume transport.

Particular attention is given to buoyancy-driven motion, ceiling accumulation, and the interaction between hydrogen transport and the ventilation system.

5.2 Global Hydrogen Dispersion Behavior

The simulations reveal that hydrogen dispersion follows a characteristic sequence of physical stages.

Immediately after release, hydrogen remains concentrated near the leak source. As time progresses, buoyancy becomes the dominant transport mechanism, causing the plume to rise rapidly toward the ceiling.

Once the ceiling is reached, hydrogen spreads horizontally and forms a concentration layer in the upper region of the enclosure. The ventilation system subsequently influences the redistribution and removal of hydrogen from the room.

This behavior is consistent with the expected physical properties of hydrogen and confirms the validity of the simulation framework.

5.3 Initial Hydrogen Release (5 s)

Figure 5.1 illustrates the hydrogen concentration field shortly after the beginning of the leak.

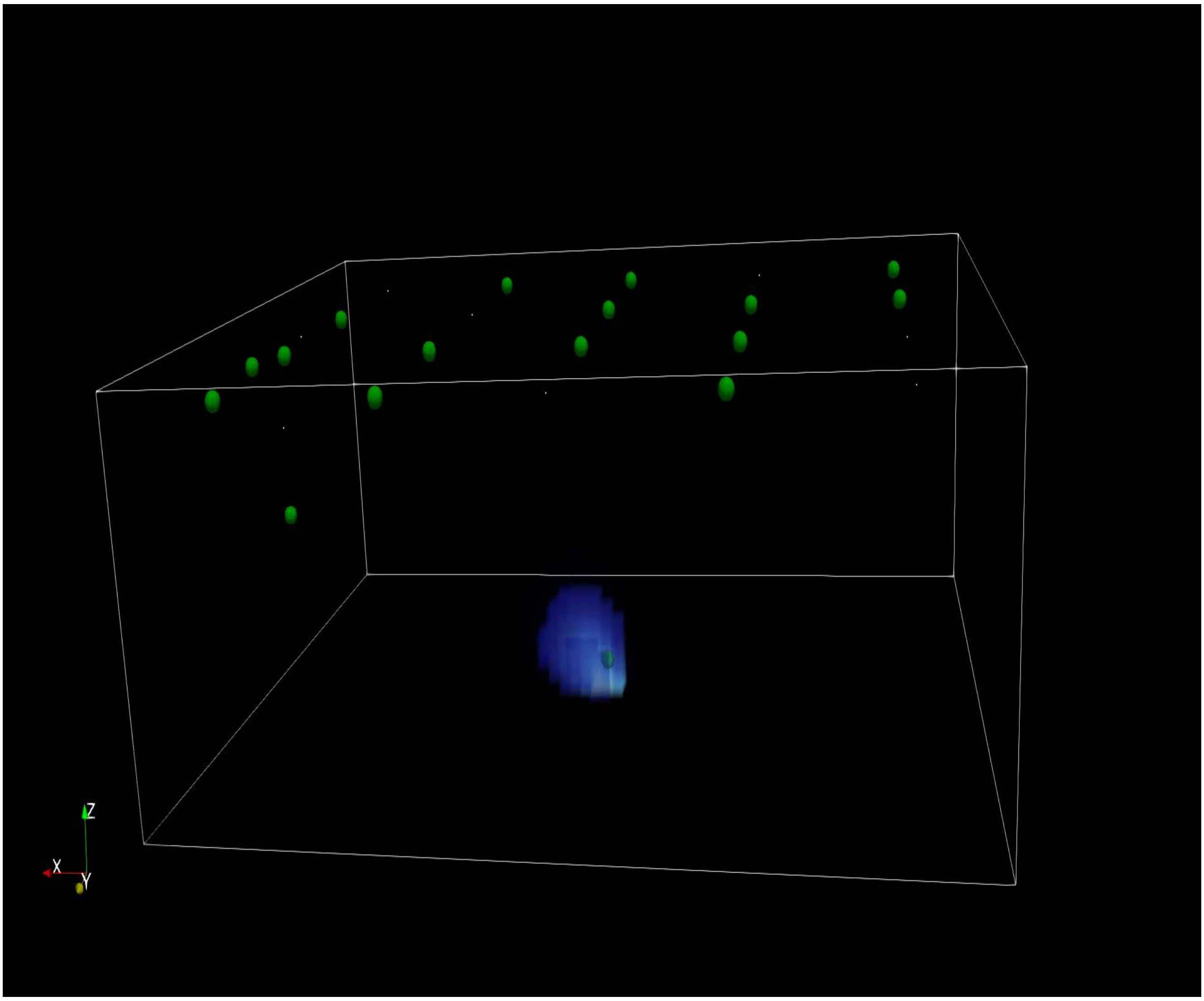

Figure 5.1: Hydrogen plume shortly after release (5 s).

At this stage, hydrogen remains concentrated around the source location. The plume occupies only a limited region of the computational domain and the influence of buoyancy is still developing.

Most sensors remain unaffected because the hydrogen cloud has not yet reached the upper part of the room.

5.4 Buoyant Rise Phase (20 s)

The hydrogen distribution after twenty seconds is shown in Figure 5.2.

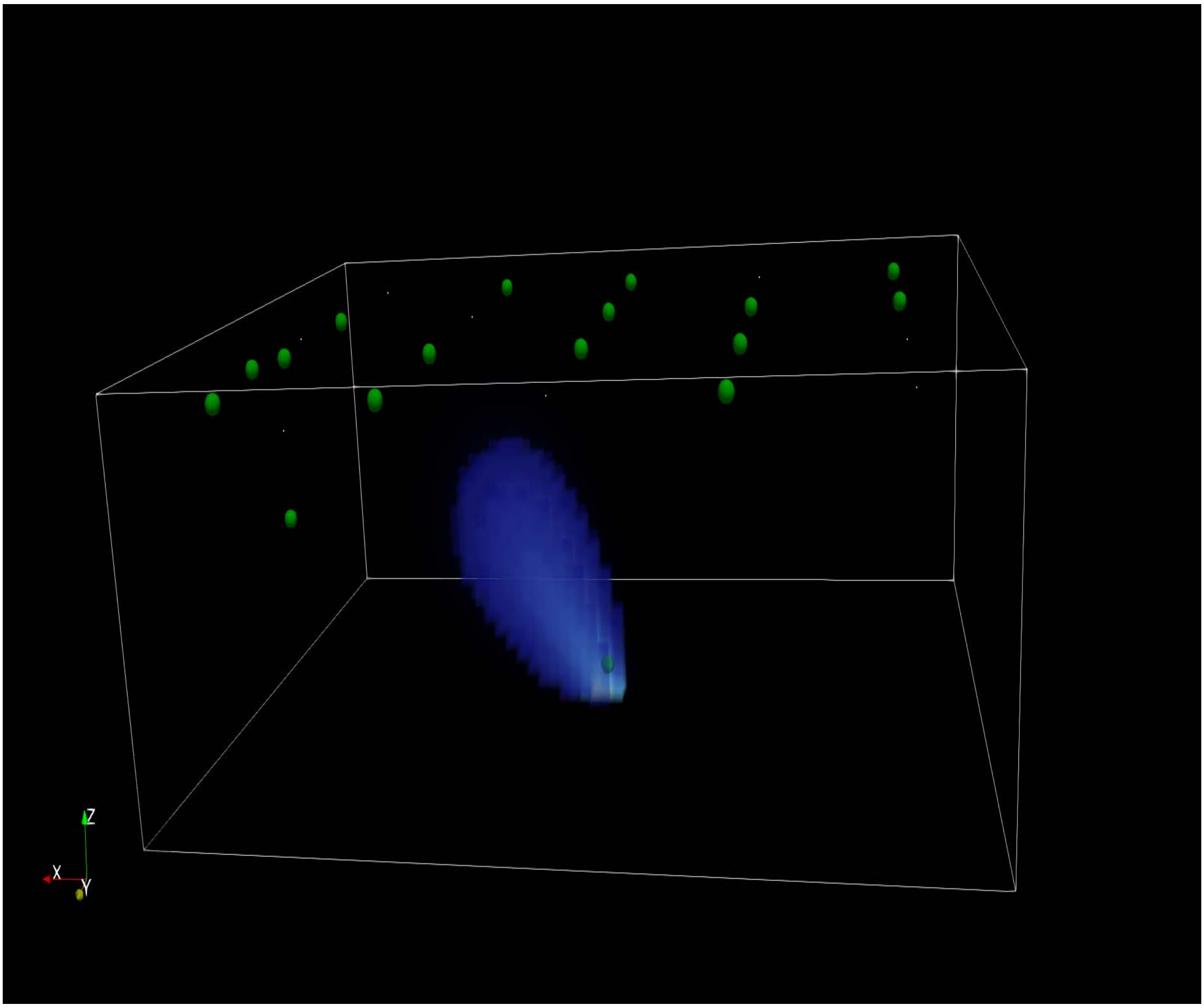

Figure 5.2: Hydrogen buoyant rise (20 s).

The plume begins to move upward rapidly due to buoyancy forces. Vertical transport becomes dominant and hydrogen starts approaching the ceiling region.

This stage demonstrates the importance of density differences between hydrogen and air. The observed motion confirms that ceiling-mounted sensors are expected to detect hydrogen earlier than sensors installed at lower elevations.

5.5 Ceiling Accumulation Phase (80 s)

The hydrogen distribution at eighty seconds is presented in Figure 5.3.

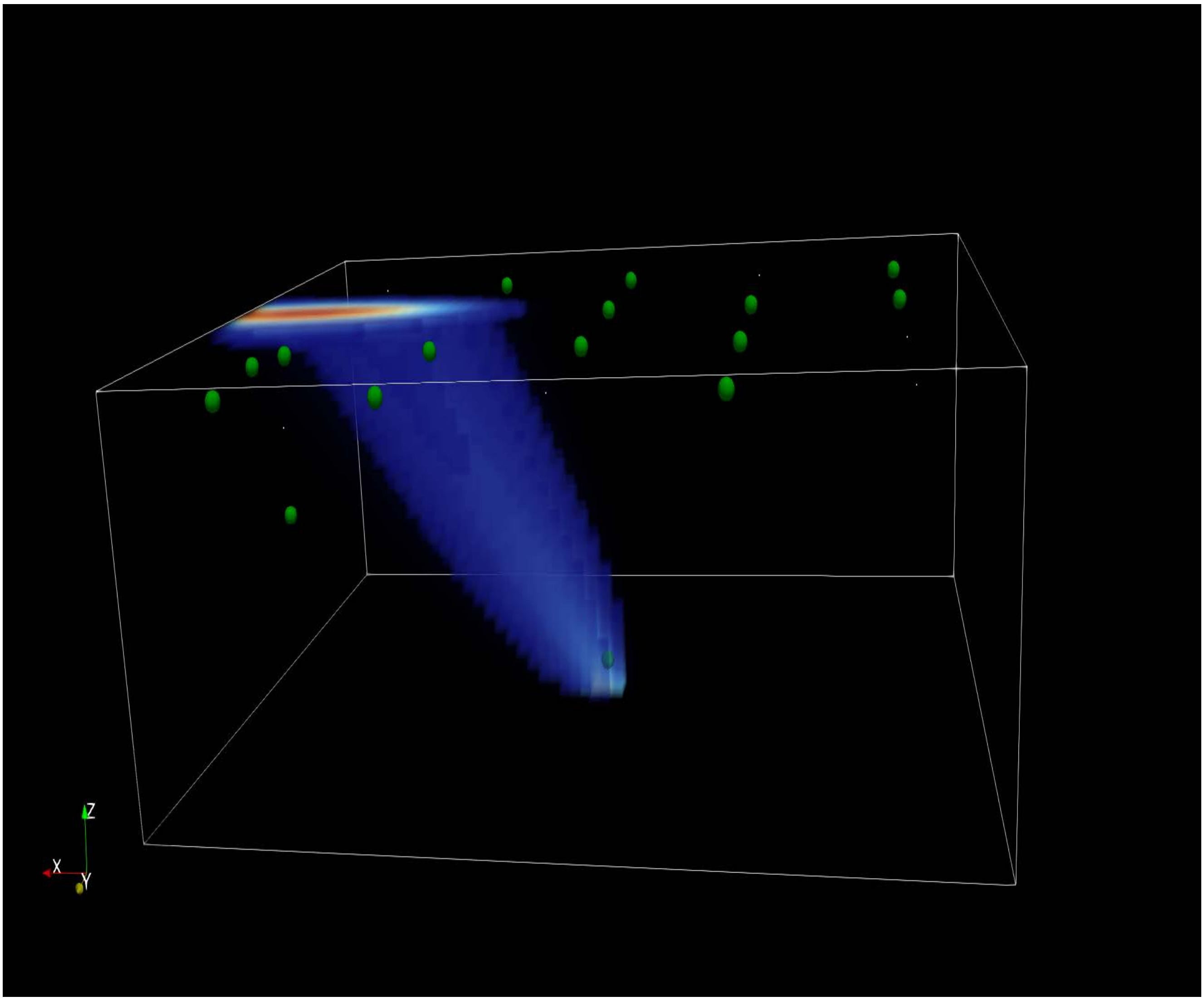

Figure 5.3: Hydrogen accumulation near the ceiling (80 s).

After reaching the ceiling, the hydrogen plume spreads horizontally across the upper region of the enclosure.

This accumulation phenomenon represents one of the most important observations of the study because it directly influences sensor performance.

The concentration field clearly demonstrates that ceiling-mounted sensors are located in the region containing the highest hydrogen concentrations.

5.6 Quasi-Steady State Regime (150 s)

The hydrogen concentration field after one hundred and fifty seconds is shown in Figure 5.4.

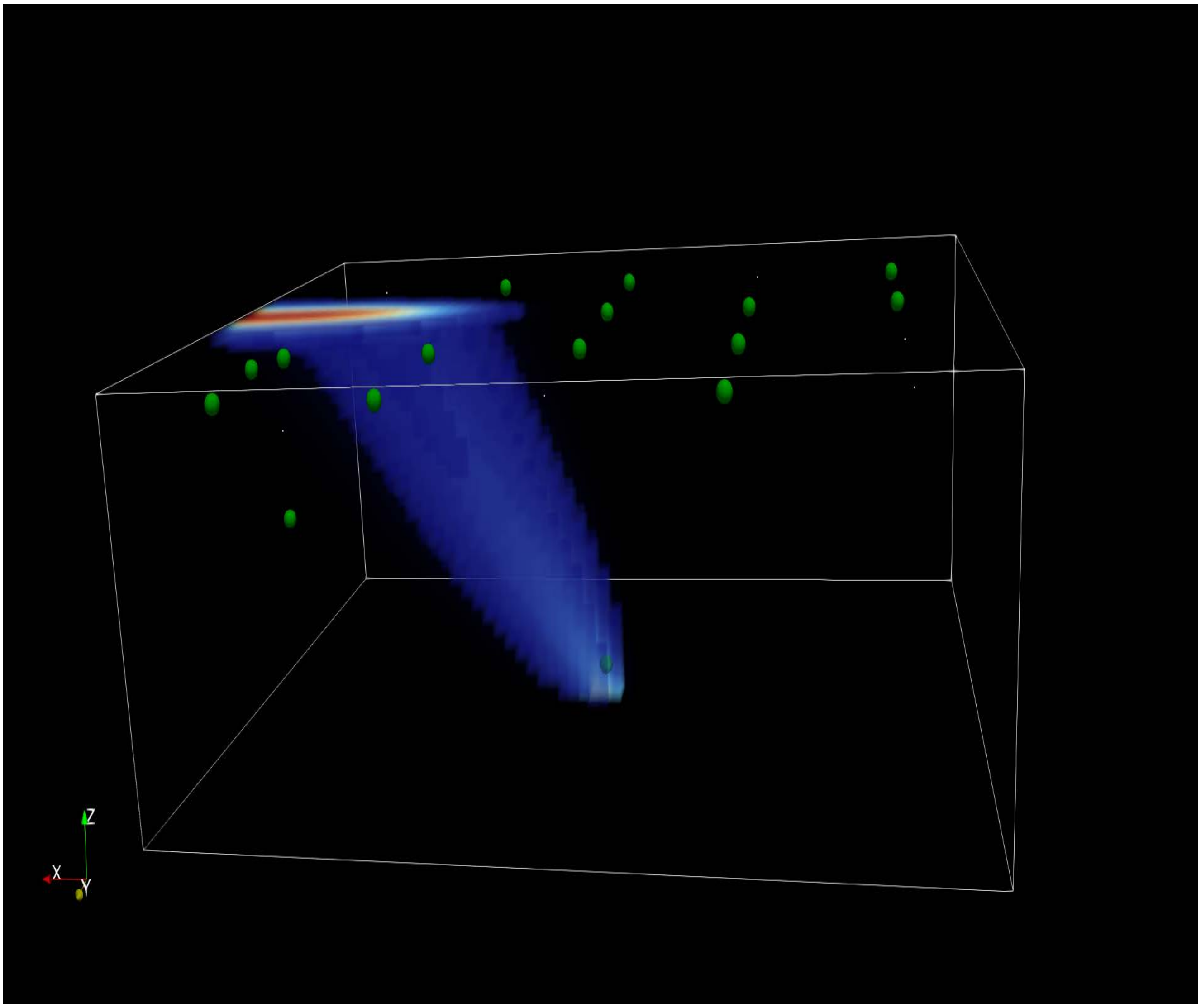

Figure 5.4: Quasi-steady hydrogen distribution (150 s).

At this stage, the system approaches a quasi-steady state. The overall structure of the hydrogen plume becomes relatively stable while ventilation continuously removes part of the gas from the room.

The concentration distribution no longer changes significantly compared with previous stages, indicating that the dominant transport mechanisms have reached a dynamic equilibrium.

5.7 Influence of Buoyancy

The simulations clearly show that buoyancy is the dominant physical mechanism governing hydrogen transport.

Unlike heavier gases, hydrogen rapidly migrates toward the ceiling after release. This behavior explains why the highest concentrations are consistently observed in the upper region of the enclosure.

The dominance of buoyancy has important implications for the design of hydrogen monitoring systems and supports the use of ceiling-mounted sensors.

5.8 Influence of Ventilation

Ventilation affects both the shape and evolution of the hydrogen plume.

The inlet airflow modifies plume trajectories, while the outlet contributes to hydrogen removal. The interaction between buoyancy and ventilation determines the final concentration distribution observed in the room.

The simulations indicate that proper ventilation can significantly reduce hydrogen accumulation and improve overall safety conditions.

5.9 Discussion

The CFD simulations successfully reproduce the expected behavior of hydrogen inside enclosed environments.

Several important observations can be highlighted:

- Hydrogen rises rapidly due to buoyancy.
- Ceiling accumulation occurs consistently across leak scenarios.
- Ceiling-mounted sensors are exposed to elevated concentrations earlier than lower sensors.
- Ventilation influences hydrogen transport and removal.
- The concentration field stabilizes after sufficient simulation time.

These observations provide the physical basis for the sensor ranking analysis and machine-learning localization framework presented in the following chapters.

5.10 Summary

The CFD results demonstrate that hydrogen dispersion inside the room is primarily controlled by buoyancy and ventilation.

The simulations reveal strong hydrogen accumulation near the ceiling, confirming the effectiveness of the selected sensor placement strategy.

These findings constitute the physical foundation upon which the Artificial Intelligence localization system is developed.

Chapter 6

Dataset Generation and Artificial Intelligence Framework

6.1 Introduction

While Computational Fluid Dynamics provides detailed information regarding hydrogen dispersion, practical leak localization requires a method capable of estimating the source position directly from sensor measurements.

Artificial Intelligence offers a powerful solution to this problem. By learning the relationship between sensor responses and leak coordinates, a machine-learning model can estimate the leak location without running additional CFD simulations.

This chapter describes the generation of the machine-learning dataset and the development of the Artificial Intelligence framework used for hydrogen leak localization.

6.2 Data Extraction from CFD Simulations

Each CFD simulation generated concentration measurements at all sensor locations throughout the simulation period.

The collected information includes:

- Hydrogen concentration values.
- Detection times.
- Maximum recorded concentrations.
- Temporal concentration evolution.
- Leak coordinates.

These measurements were automatically extracted and organized into structured datasets suitable for machine-learning applications.

6.3 Generated Dataset

The final database was generated from all simulated leak scenarios.

The resulting dataset contains information from:

- 106 independent leak scenarios.
- 25 monitoring sensors.
- Approximately 1.59 million measurements.
- 30 machine-learning features.

The dataset represents a comprehensive collection of hydrogen dispersion patterns covering multiple leak locations and release conditions.

6.4 Dataset Statistics

Table 6.1 summarizes the main characteristics of the generated database.

Parameter	Value
Leak Scenarios	106
Number of Sensors	25
Sensor Measurements	1,590,000
Feature Count	30
AI Model	Random Forest

Table 6.1: Summary of the generated AI dataset.

The large number of measurements provides sufficient information for training and evaluating machine-learning models.

6.5 Feature Engineering

Raw sensor measurements were transformed into numerical features suitable for machine-learning analysis.

The extracted features include:

- Maximum concentration.
- Mean concentration.
- Detection time.
- Concentration gradients.
- Sensor response statistics.

These features summarize the information contained in the original time-series data and reduce computational complexity.

6.6 Machine Learning Workflow

The Artificial Intelligence workflow developed in this study follows several sequential steps:

1. CFD simulation.
2. Sensor data extraction.
3. Dataset generation.
4. Feature engineering.
5. Model training.
6. Performance evaluation.
7. Leak localization prediction.

The workflow establishes a direct connection between physics-based simulations and data-driven leak localization.

6.7 Random Forest Regression

A Random Forest Regressor was selected for the leak localization task.

Random Forest is an ensemble learning method based on multiple decision trees. It is particularly suitable for nonlinear regression problems and provides robust predictions even when relationships between variables are complex.

Another important advantage of Random Forest is its ability to estimate feature importance, allowing identification of the most informative sensors.

6.8 Training and Validation

The generated dataset was divided into training and testing subsets.

The training subset was used to construct the Random Forest model, while the testing subset was used to evaluate prediction performance.

The objective of the model is to estimate the leak coordinates using only sensor measurements.

6.8.1 Training and Testing Sets

The generated dataset was randomly divided into training and testing subsets.

Eighty percent of the samples were used for model training while twenty percent were reserved for performance evaluation.

This strategy ensures that the reported localization accuracy reflects the model's ability to generalize to unseen leak scenarios.

6.9 Performance Metrics

Two standard metrics were used to evaluate localization accuracy:

6.9.1 Mean Absolute Error

The Mean Absolute Error (MAE) is defined as:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y_i}| \tag{6.1}$$

where y_i represents the true value and $\hat{y_i}$ represents the predicted value.

6.9.2 Root Mean Square Error

The Root Mean Square Error (RMSE) is defined as:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y_i})^2} \tag{6.2}$$

RMSE penalizes large prediction errors more strongly than MAE and is therefore useful for evaluating localization performance.

6.10 Localization Results

The trained Random Forest model achieved the following performance:

Axis	MAE (m)	RMSE (m)
X	0.572	0.650
Y	0.107	0.150
Z	0.000	0.000

Table 6.2: Random Forest localization performance.

The results indicate that the model successfully captures the relationship between sensor measurements and leak coordinates.

The largest prediction uncertainty is observed along the x-axis, while the y-axis predictions remain highly accurate.

The zero error obtained along the z-axis does not indicate perfect prediction capability. Instead, it results from the fact that all simulated leak sources were located at the same vertical position throughout the simulation campaign.

The zero error along the z-axis is explained by the fact that most leak sources were located at the same vertical level during the simulations.

6.11 Overall Localization Workflow

The complete workflow developed in this project is illustrated below.

CFD Simulations
↓
Sensor Network Measurements
↓
Dataset Generation
↓
Random Forest Training
↓
Leak Localization
↓
Future Autonomous Robot

Figure 6.1: Overall CFD-AI localization workflow.

6.12 Summary

This chapter presented the complete Artificial Intelligence framework developed for hydrogen leak localization.

The generated dataset contains more than one million sensor measurements obtained from 106 CFD simulations. A Random Forest model was successfully trained to estimate leak positions from sensor data and achieved promising localization accuracy.

The following chapter investigates sensor ranking and identifies the most informative sensor locations for hydrogen leak detection and localization.

Chapter 7

Sensor Ranking Analysis

7.1 Introduction

The performance of a hydrogen monitoring system strongly depends on sensor placement. Although all sensors provide useful information, some locations are significantly more informative than others for leak detection and localization.

This chapter evaluates the performance of the sensor network and identifies the most influential sensors using both physical and Artificial Intelligence criteria.

7.2 Evaluation Metrics

Several performance indicators were used to assess sensor effectiveness:

- Detection coverage.
- Average detection time.
- Median detection time.
- First detection frequency.
- Maximum recorded concentration.
- Distance from leak source.

These metrics provide complementary information regarding sensor performance.

7.3 CFD-Based Sensor Ranking

The ranking obtained from the CFD analysis identified several high-performance sensors.

The best-performing sensors are presented in Table 7.1.

Sensor ID	Relative Performance
S19_ceiling_grid	Excellent
S04_ceiling_grid	Excellent
S13_ceiling_grid	Very Good
S20_ceiling_grid	Very Good
S18_ceiling_grid	Very Good

Table 7.1: Highest ranked sensors based on CFD analysis.

These sensors consistently detected hydrogen early and exhibited high coverage across multiple leak scenarios.

7.4 Artificial Intelligence Feature Importance

The Random Forest model provides feature importance values that indicate which sensors contribute most strongly to leak localization.

The most influential sensors identified by the AI model are:

Sensor ID	Importance
S08_ceiling_grid	0.191
S20_ceiling_grid	0.187
S05_ceiling_grid	0.112
S13_ceiling_grid	0.089
S19_ceiling_grid	0.059

Table 7.2: Most important sensors identified by the Random Forest model.

7.5 Comparison Between CFD and AI Results

A remarkable agreement is observed between the CFD sensor ranking and the Artificial Intelligence feature importance analysis.

Several sensors appear in both rankings, particularly:

- S19_ceiling_grid
- S20_ceiling_grid
- S13_ceiling_grid

This consistency confirms that the machine-learning model correctly captures the underlying physical behavior observed in the simulations.

7.6 Physical Interpretation

The dominance of ceiling-mounted sensors can be explained directly by hydrogen physics.

Because hydrogen is significantly lighter than air, buoyancy causes the plume to rise rapidly after release. Consequently, the highest concentrations are generally observed near the ceiling.

Sensors installed in this region are therefore exposed to hydrogen earlier and receive more informative concentration signals.

This observation explains both the CFD ranking results and the Artificial Intelligence feature importance analysis.

7.7 Implications for Sensor Network Design

The results suggest that future hydrogen monitoring systems should prioritize ceiling-mounted sensors.

A reduced network containing only the most informative sensors could potentially achieve localization performance comparable to the complete network while reducing installation costs.

Such optimization represents an important direction for future work.

7.8 Summary

The sensor ranking analysis demonstrates that ceiling-mounted sensors provide the highest information content for hydrogen leak localization.

The strong agreement between CFD-based rankings and Artificial Intelligence feature importance confirms the validity of the proposed methodology and highlights the importance of physics-informed sensor placement strategies.

Chapter 8

Future Autonomous Robotic Inspection System

8.1 Introduction

The ultimate objective of hydrogen leak localization is not only to detect leaks but also to enable rapid and autonomous intervention.

The CFD simulations and Artificial Intelligence framework developed in this project provide the foundation for a future robotic inspection system capable of identifying and locating hydrogen leaks in real time.

This chapter presents a conceptual architecture for such a system and discusses potential future developments.

8.2 System Architecture

The proposed robotic inspection platform combines fixed sensors, Artificial Intelligence, and a mobile robotic unit.

The overall workflow can be summarized as:

Leak Source $\rightarrow$ Sensor Network $\rightarrow$ AI Localization $\rightarrow$ Mobile Robot $\rightarrow$ Leak Confirmation

The sensor network continuously monitors the environment and provides concentration measurements to the Artificial Intelligence module.

The AI system estimates the probable leak coordinates and transmits this information to the mobile robot.

The robot then navigates toward the predicted location and performs additional verification measurements.

8.3 Fixed Sensor Network

The first layer of the system consists of the fixed sensor network developed in this project.

The sensor network provides:

- Continuous monitoring.
- Early leak detection.
- Concentration mapping.
- Initial leak localization.

The CFD and AI results demonstrated that ceiling-mounted sensors provide the highest information content and therefore constitute the most important component of the monitoring system.

8.4 Artificial Intelligence Module

The Artificial Intelligence module acts as the decision-making component of the system.

Its primary functions include:

- Processing sensor measurements.
- Estimating leak coordinates.
- Evaluating localization confidence.
- Guiding robotic navigation.

The Random Forest model developed in this study represents the first implementation of this concept.

Future versions could incorporate advanced deep-learning models and real-time prediction capabilities.

8.5 Mobile Robotic Platform

A future robotic platform could be equipped with:

- Hydrogen sensors.
- Cameras.
- LIDAR systems.
- Wireless communication modules.
- Autonomous navigation capabilities.

The robot would receive leak position estimates from the AI system and move toward the predicted location.

Once the leak region is reached, onboard sensors could perform additional measurements to confirm the source position.

8.6 Autonomous Navigation

Autonomous navigation is a key component of the proposed system.

The navigation process could involve:

1. Receiving predicted leak coordinates.
2. Generating a collision-free path.
3. Moving toward the target location.
4. Continuously updating localization estimates.
5. Confirming the leak source.

Such a strategy would significantly reduce human exposure to potentially hazardous environments.

8.7 Real-Time Digital Twin

An advanced version of the system could integrate a digital twin of the monitored environment.

The digital twin would continuously combine:

- Sensor measurements.
- CFD predictions.
- AI localization results.
- Robot observations.

This integrated framework would provide a real-time representation of hydrogen dispersion and support intelligent safety management.

8.8 Industrial Applications

Potential applications include:

- Hydrogen storage facilities.
- Fuel-cell stations.
- Chemical plants.
- Energy infrastructures.
- Research laboratories.
- Industrial safety monitoring systems.

The proposed framework is not limited to hydrogen and could potentially be adapted to other gases and industrial monitoring applications.

8.9 Summary

The CFD simulations, sensor network design, and Artificial Intelligence framework developed in this project establish the foundation for a future autonomous hydrogen leak inspection system.

The integration of fixed sensors, machine learning, and mobile robotics represents a promising direction for intelligent industrial safety systems.

Chapter 9

Conclusion and Future Work

9.1 Project Overview

This project presented a complete computational framework for hydrogen leak localization inside a three-dimensional indoor environment.

The proposed methodology combines Computational Fluid Dynamics, sensor network analysis, and Artificial Intelligence to detect and localize hydrogen leaks under a wide range of operating conditions.

The study demonstrates how physics-based simulations can be integrated with data-driven methods to support the development of intelligent safety monitoring systems.

9.2 Main Achievements

Several important objectives were successfully achieved during this project.

First, a three-dimensional CFD environment was developed using OpenFOAM to simulate hydrogen dispersion inside an enclosed ventilated room.

Second, a distributed sensor network composed of twenty-five monitoring locations was designed and implemented.

Third, a large simulation campaign was performed, generating a total of 106 independent leak scenarios representing different leak positions and release conditions.

Fourth, a complete database was automatically generated from the CFD simulations, resulting in approximately 1.59 million sensor measurements suitable for Artificial Intelligence applications.

Finally, a Random Forest regression model was successfully trained to estimate hydrogen leak coordinates directly from sensor measurements.

9.3 Key Scientific Findings

The CFD simulations revealed several important physical observations.

Hydrogen dispersion is dominated by buoyancy due to the large density difference between hydrogen and air.

Following release, hydrogen rapidly rises toward the ceiling and forms an accumulation layer in the upper region of the enclosure.

The simulations further demonstrated that ceiling-mounted sensors consistently provide earlier detection and more informative measurements than sensors located near the floor.

The machine-learning analysis confirmed these observations by identifying ceiling sensors as the most influential features for leak localization.

A strong agreement was therefore observed between the physical interpretation of the CFD results and the Artificial Intelligence feature importance analysis.

9.4 Artificial Intelligence Performance

The Artificial Intelligence framework successfully learned the relationship between sensor responses and leak coordinates.

The Random Forest model achieved:

- $\text{MAE}_x = 0.572$ m
- $\text{RMSE}_x = 0.650$ m
- $\text{MAE}_y = 0.107$ m
- $\text{RMSE}_y = 0.150$ m

These results demonstrate the feasibility of AI-based hydrogen leak localization using sensor measurements generated from CFD simulations.

The obtained accuracy is promising and confirms the potential of machine learning for industrial safety applications.

9.5 Project Contributions

The main contributions of this work can be summarized as follows:

- Development of a complete CFD hydrogen dispersion framework.
- Simulation of 106 hydrogen leak scenarios.
- Design of a 25-sensor monitoring network.
- Generation of a large-scale AI-ready dataset.
- Development of a machine-learning leak localization model.
- Identification of the most informative sensor locations.
- Proposal of a future autonomous robotic inspection architecture.

These contributions establish a bridge between Computational Fluid Dynamics, Artificial Intelligence, and intelligent safety monitoring systems.

9.6 Limitations of the Study

Although the results are encouraging, several limitations remain.

The current study is based entirely on numerical simulations and does not yet include experimental validation.

The leak sources were also generated under controlled conditions and do not represent all possible industrial scenarios.

Furthermore, the Artificial Intelligence model was trained using synthetic CFD-generated data and has not yet been tested using measurements obtained from real hydrogen sensors.

These limitations should be addressed in future studies.

9.7 Future Work

Several directions can be explored to extend the present work.

Future developments may include:

- Experimental validation using real hydrogen sensors.
- Additional leak scenarios with different source heights.
- More complex industrial geometries.
- Investigation of advanced machine-learning models.
- Deep-learning approaches for leak localization.
- Real-time data processing.
- Sensor network optimization.
- Integration with mobile robotic platforms.
- Development of a digital twin framework.

These improvements would further increase the practical applicability of the proposed methodology.

9.8 Final Remarks

Hydrogen is expected to play an increasingly important role in future energy systems. Ensuring its safe deployment requires innovative monitoring and localization technologies capable of rapidly identifying accidental leaks.

The present work demonstrates that the combination of Computational Fluid Dynamics, sensor networks, and Artificial Intelligence provides a powerful approach for hydrogen leak localization.

Beyond the results obtained in this study, the developed framework establishes a foundation for future intelligent safety systems capable of combining simulation, sensing, machine learning, and autonomous robotics within a unified platform.

Such technologies may contribute significantly to the safe and efficient deployment of hydrogen infrastructures in the coming decades.

References

1. OpenFOAM Foundation. OpenFOAM User Guide.
2. ParaView Documentation.
3. Breiman, L. Random Forests. Machine Learning, 2001.
4. Molkov, V. Fundamentals of Hydrogen Safety Engineering.
5. Crowl, D. A., Louvar, J. F. Chemical Process Safety.
6. Turns, S. R. An Introduction to Combustion.
7. Versteeg, H., Malalasekera, W. Computational Fluid Dynamics.
8. Bishop, C. M. Pattern Recognition and Machine Learning.
9. Goodfellow, I., Bengio, Y., Courville, A. Deep Learning.
10. Scikit-Learn Documentation.

Appendix A

GitHub Repository

The complete project source code, OpenFOAM cases, Python analysis scripts, and datasets are available at:

https://github.com/mhamadcheikh416-creator/HydrogenLeak-CFD-AI-Localization

Appendix B

Additional Results

This appendix contains supplementary figures, sensor statistics, and additional CFD visualizations that support the main conclusions of the study.